

Mistri

A WhatsApp assistant that answers a garage's customers on the number they already message.

Product	Mistri — WhatsApp assistant for car garages
The name	What half of Dubai already calls the mechanic
First customer	Care Auto Repair Services, Dubai (careautorepair.ae)
Version	v1 — in build
Document date	1 September 2026
Owner	Noman

What this document is

A plain description of what the tool does, what it will never do, and how it fits into a garage's day. Sections 1–4 are written so a garage owner can read them. Sections 5–9 are the build detail.

01

What this tool is

Mistri answers a garage's WhatsApp messages automatically. It runs on the garage's **existing WhatsApp Business number** — the same number already on the shop sign, the Google listing and every old customer's phone. Nothing new to print, nothing for customers to install, nothing for staff to learn.

The owner keeps using WhatsApp Business on his phone exactly as before. The bot sits behind the same number and picks up the messages he has not answered yet. The moment he types a reply himself, the bot goes quiet in that chat.

The problem it solves

A busy garage loses work in the gaps. A customer asks “how much for brake pads?” at 9 pm, gets no answer until 10 am, and by then has booked somewhere else. The same six questions arrive twenty times a day and eat the owner's attention while he is under a car.

- **After-hours messages die.** Nobody is at the phone from 8 pm to 8 am, and that is when people sit down and think about their car.
- **The same questions, endlessly.** Price, timing, location, do you take card, do you collect the car.
- **Price questions go cold.** A quote that takes two hours to arrive is a quote the customer has stopped waiting for.
- **Nothing is measured.** No garage owner can tell you how many WhatsApp enquiries he got last month, or how many turned into work.

What it does about it

It answers in seconds, in the customer's own language, at 3 am on a Friday, using nothing but the garage's own price sheet — and it books the car in.

The one-line version

Every enquiry gets an answer within seconds, every answer is a real price off the garage's own sheet, and anything the bot is not certain about goes straight to a human.

02

The four rules

Most “AI for business” tools fail on trust: the bot invents a price, or guesses what is wrong with a car, and the garage spends a week undoing the damage. These four rules are not settings. They are the product, and they are enforced in code and in tests.

Rule 1 — Never invent a price

The bot quotes only what is written on the garage's own price sheet, word for word, fixed price or range. It never estimates, never interpolates between two rows, never guesses from a similar car. If the service or the car is not on the sheet, it offers a free inspection and alerts the owner.

Rule 2 — Never diagnose

“There's a noise”, “the light is on”, “it smells of burning”, “it won't start” — the bot does not guess a cause and does not guess a price. It collects the make, model and year plus the symptom, and books an inspection. A garage that diagnoses over WhatsApp is a garage with an argument coming.

Rule 3 — Hand over when unsure

Anything outside the price sheet and the FAQ gets one honest line — “our service advisor will call you shortly” — and the owner receives a WhatsApp alert with a three-line summary and the customer's number. The bot then stops replying in that conversation until it is released. No bluffing.

Rule 4 — Speak the customer's language

English, Arabic, Hindi and Urdu, in whichever script the customer used — including Roman Urdu and Roman Hindi, which is how most of Dubai actually types. The reply comes back in the same language and the same script.

How this is proved, not promised

A test set of 100 questions ships with the product — including 20 trick questions asking for prices that are deliberately not on the sheet, and 20 symptom messages fishing for a diagnosis. The build is not finished until that set produces **zero invented prices and zero diagnoses**.

03

Features

Customer-facing

Feature	What happens
Price answers	Identifies the service and the car, asks for make / model / year if missing, and quotes the garage's sheet price for that car category — sedan, SUV or luxury.
Booking	Collects name, car, service and preferred day, checks the garage's real opening hours and how many cars it can take that hour, confirms the slot and sends a maps link.
Hours, location, payment, warranty, pickup	Answered instantly from the garage's own FAQ. No more typing the address forty times a week.
Symptom messages	Collects the car and the symptom, offers a free inspection, books it. Never names a cause, never names a price.
Reminders	One the day before the appointment, one on the morning of it. Fewer no-shows, which is where a garage quietly loses a day.
Four languages	English, Arabic, Hindi, Urdu — script or Roman, matched to whatever the customer wrote in.
Short replies	One to four lines. One question at a time. It reads like a person on WhatsApp, not a brochure.

Owner-facing

Feature	What happens
Booking alert	A WhatsApp message to the owner's personal number the moment a car is booked in.
Handoff alert	A three-line summary plus the customer's number whenever the bot steps back.
Daily 6 pm summary	Conversations, bookings, handoffs, and how many messages arrived after closing.
Takeover	He replies from his own phone and the bot goes silent in that chat for two hours.
Release	/bot on 0501234567 from his phone hands the chat back.
Admin page	A password-protected page listing conversations, full transcripts and bookings. Deliberately plain.

The pilot report

One command prints the numbers for any date range: conversations and how many started outside opening hours, price questions answered, bookings, handoffs, median reply time, language split.

This is not a vanity dashboard. It is the evidence that sells garage number two: not “AI will transform your business”, but “here is what happened at Care over thirty days.”

04

How it works on the owner's phone

This is the part that makes the product sellable. Understand it before promising anything to a garage.

Coexistence — same number, two places

One number can now run on the WhatsApp Business app **and** the official API at the same time — Meta calls it coexistence. The owner keeps his app, his chats and his history.

- Nothing installs on anyone's phone — not the owner's, not the customer's.
- The garage keeps its number, its chat history and every existing contact.
- Setup is one QR scan by the owner, roughly twenty minutes, done once.
- It can be disconnected again at any time. The number simply goes back to being app-only.

The handover, in practice

Situation	What the bot does
Customer asks a price that is on the sheet	Answers it, then offers the next free slots
Customer asks something off the sheet	Offers a free inspection, alerts the owner, stops replying
Owner types a reply himself	Goes silent in that chat for 2 hours
Owner sends /bot on <number>	Starts replying in that chat again
Handed-over chat goes quiet for 24 hours	Releases itself automatically

The bot never talks over the owner. That is the whole design of this section.

What the owner must be told before he agrees

He should hear this before the QR scan, not after. Full list in `ONBOARDING.md`.

Stops working on that number	Keeps working normally
Broadcast lists — no new ones, existing ones become read-only	Voice and video calls in WhatsApp
Disappearing messages and view-once	Normal phone calls — same SIM, unaffected
Live location in one-to-one chats	Group chats (the bot never sees or replies in them)
WhatsApp Web is unlinked once at setup; re-link after	Status, catalog and orders

Say the broadcast-list one out loud. It is the only change a garage might genuinely feel, and some owners send promos that way.

05

How it is built

A single Python service on a VPS. No queue, no cluster, no Kubernetes — one garage's WhatsApp traffic is a few hundred messages a day.

Layer	Choice	Why
Service	Python 3.11, FastAPI, uvicorn	WhatsApp pushes webhooks; this handles them and acks fast
Hosting	AlmaLinux VPS, systemd, nginx, Let's Encrypt	Already owned. Meta requires a public HTTPS endpoint
WhatsApp	Meta Cloud API, via 360dialog for the pilot	Coexistence needs an approved provider — see section 7
Language model	Anthropic API — Sonnet first, Haiku once stable	Reply quality first, then benchmark for cost
Database	SQLite via SQLAlchemy	One file, no server. Moving to Postgres is a connection string
Knowledge	YAML files per garage	The owner's price sheet is small. No vector database, no RAG
Scheduling	APScheduler	Reminders and the 6 pm summary

Two decisions worth knowing

Multi-garage from day one

Every table and every config carries a `garage_id`. Garage number two is a new folder of YAML files and a number connected — not a fork of the codebase. This costs almost nothing now and would cost weeks later.

One WhatsApp module, nothing else

All WhatsApp traffic goes through a single file. Switching from the pilot provider to Meta direct is two environment variables. No other part of the code knows or cares which one is in use.

Knowledge files, per garage

File	Holds
<code>prices.yaml</code>	The price sheet — every service by car category, or per model where the garage prices that way. The bot may quote nothing that is not in this file.
<code>faq.yaml</code>	The answers that are not prices, built from the garage's own chat history
<code>info.yaml</code>	Hours, address, maps link, payment methods, warranty, pickup, capacity per hour, owner's alert number

The owner's prices live in a plain text file he could read himself. That is deliberate — when a garage asks “where does it get the price from?”, the honest answer is a file you can open in front of him.

06

What it will not do

Written down so it stops being discussed. These are out of scope for v1 by decision, not by oversight.

Not in v1	Reason
Full web dashboard	v1 reporting is WhatsApp alerts plus a bare admin page. A dashboard is a product of its own
Payments and deposits	Adds a payment provider, refunds and disputes to a two-week build
CRM and dealer-software integration	Every garage uses something different. Wait for a customer who will pay for it
Mobile app	The whole point is that nothing installs
Voice or AI phone calls	Different product, different failure modes
Marketing broadcasts	Coexistence removes broadcast lists anyway. A real v2 feature, not a v1 bolt-on
Multi-branch	One garage, one location, until a customer has two
Vector search over the price sheet	The sheet fits in the prompt. Adding retrieval would add a way to get prices wrong

Ideas that come up mid-build are written to `LATER.md` and dropped. The v1 that ships beats the v2 that is still being designed.

Definition of done

v1 is finished when all three of these are true — not before, and not on any other basis.

- 1

Care's real number answers "brake pads Camry 2019?" with the sheet price in under five seconds, offers slots, books one, and the owner gets the alert.
- 2

The 100-question test set — 20 of them fishing for off-sheet prices, 20 fishing for a diagnosis — produces zero invented prices and zero diagnoses.
- 3

The report command prints correct numbers for the pilot period.

07

Getting a garage connected

Coexistence cannot be switched on by the garage alone. Meta only allows it through an approved provider's onboarding flow, and the owner must click through it himself — nobody may do it on his behalf.

The route chosen for the pilot

	Via a provider (360dialog)	Direct with Meta
Meta app review needed	No	Yes — advanced access on two permissions
Business verification needed	No	Yes
Time before the pilot can go live	Days	Weeks, in a queue you do not control
Monthly cost	~€49 plus Meta's per-conversation fees	Meta's fees only

Decision

Pilot number one goes through 360dialog. Meta's own approval queue would consume the entire build window, and switching to direct later is a two-line configuration change — not a rewrite.

What the owner does, once

- 1 Updates WhatsApp Business to 2.24.17 or newer (the day before — an old app is the usual reason this fails)
- 2 Signs in with his own Meta business account and picks “connect your existing WhatsApp Business app”
- 3 Re-enters and verifies his number
- 4 Opens the message that arrives in his WhatsApp Business app and scans the QR code
- 5 Chooses to sync chat history — say yes; it is where the FAQ comes from
- 6 Re-links WhatsApp Web afterwards if he uses it

About twenty minutes, done once, with someone sitting next to him. The full script, including what to do when a step fails, is in `ONBOARDING.md`.

08

Build status

Everything that does not need the garage's own data is finished, with 296 tests behind it.

Stage	Scope	Status
Step 0	Onboarding route decided; owner checklist written; message templates drafted	Done
Days 1–2	Webhook, database, message logging, owner-takeover pause	Done
Days 3–5	Reply engine with the four rules, the price guard, language handling	Done
Days 6–8	Booking flow, slot capacity, confirmation, owner alerts	Done
Days 9–10	Handoff, owner commands, reminders, the 6 pm summary	Done
Days 11–12	Admin page, pilot report, systemd and nginx deployment	Done
v1.1	Post-service follow-up and the service-due nudge	Done
Review	Two adversarial passes over the code; nine real defects fixed	Done
Days 13–14	The price sheet, their real questions, the number connected	Needs the garage

What is still needed from the garage

- **The price sheet.** Their own service list is already loaded from their website; every price is deliberately blank. Nothing published, nothing guessed.
- **The top thirty questions** from their real WhatsApp history. These become the FAQ and the test set. Invented questions produce a bot that passes its own exam and fails on day one.
- **Four message templates approved by Meta.** Reminders cannot send until they are, and it is the only step whose speed nobody here controls.

What the review found

Two passes were made over the finished code, reading it rather than running it. Nine real defects turned up while every test was passing, all of them in the plumbing around the rules rather than in the rules themselves.

- The bot would have **talked over the owner** — his replies were matched against a phone number formatted for humans, so each was filed as a customer message.
- The webhook was **open to anyone who found the URL** on the pilot's provider, which does not sign requests. A stranger could have sent messages from the garage's number.
- Several failures reached the customer as **silence** rather than as a person.

The risk that is still open

The takeover behaviour can only be fully confirmed on a live number. It is the first thing checked on connection day, and it is checked first because it was already found broken once. A bot that talks over the owner is worse than no bot.

09

Beyond the first garage

The pilot exists to produce a number, not a testimonial. Thirty days at Care should yield a line like: *“214 enquiries, 61 of them after closing time, 38 cars booked, average reply eight seconds.”* That sentence is what sells garage number two.

What garage two actually costs

Step	Effort
New folder with that garage's prices, FAQ and details	An afternoon, mostly transcription
Owner connects his number	20 minutes, once
Code changes	None — <code>garage_id</code> is already everywhere

Sensible next steps, in order

- **The service-due nudge is already built.** Months after a car's last visit it asks whether they want to come in. This is what a garage renews for: cancelling stops the thing that refills the bay.
- Owner-side reporting he can read himself, once there is enough data to be worth reading.
- Template-based promotions, which become the honest replacement for the broadcast lists coexistence removes.
- Other trades with the same shape — fixed price list, appointment, one busy owner on WhatsApp.

The tool is not interesting because it uses AI. It is interesting because a garage owner under a car at 8 pm stops losing the customer who messaged at 7:40 — and because the bot would rather say “someone will call you” than make a price up.